WEEK 3 · LECTURE 3

Self-Attention and the Transformer Decoder

How every token gathers what it needs from the others

BSc Data Science and Artificial Intelligence · Al-Zaytoonah University of Jordan · Dr. Akram Fatayer · 2026/2027

TODAY

By the end of this lecture you can…

01

Explain why a token’s meaning depends on the tokens around it

02

Compute scaled dot-product attention by hand from queries, keys and values

03

Explain the causal mask, multi-head attention and position information

04

Trace one decoder block and say where its parameters are

LAST WEEK'S RESULT

The same token "bank" got one static vector but different contextual ones

1.00

Static vectors: identical in every sentence, as they must be

0.45

Contextual vectors: the river bank vs. the bank that approved a loan

Something between the embedding table and the output mixed in the neighbours. Today: that mechanism.

THE PROBLEM

To read one word, a model has to look at the others

The animal didn’t cross the street because **it** was too **tired**.

it = the animal

The animal didn’t cross the street because **it** was too **wide**.

it = the street

لم يعبر الحيوانُ الشارعَ لأنه كان متعباً The same in Arabic: the pronoun attached to “لأنه” points back to the animal.

One changed word, a different meaning for “it”. Each token needs a way to ask the others a question and collect their answers.

QUERIES, KEYS, VALUES

Every token plays three roles at once

`q = x W_Q`

Query

What am I looking for? “it” looks for a noun it can refer to.

`k = x W_K`

Key

What do I offer? “animal” advertises: a singular noun, a living thing.

`v = x W_V`

Value

What do I hand over? The information copied into “it” if it attends to me.

A query is compared with every key. The better they match, the more of that token’s value flows in. W_Q, W_K and W_V are learned in training.

SCALED DOT-PRODUCT ATTENTION

The whole mechanism fits in one line

$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^{T} / \sqrt{d} \right) V$$

STEP 1 · QK^{T}

Score

Dot product of each query with every key

STEP 2 · ÷ √d

Scale

Divide by the square root of the key size

STEP 3 · softmax

Normalise

Turn each row of scores into weights that sum to 1

STEP 4 · × V

Mix

Each output is the weighted average of the values

BY HAND

“book” asks, and “Ali” answers most strongly

Key token	q · k	÷ √3	weight
Ali	5	2.89	**0.75**
reads	0	0.00	0.04
a	1	0.58	0.07
book	2	1.15	0.13

```
q(book) = [2, 0, 1]
k(Ali) = [2, 0, 1]
q · k = 2·2 + 0·0 + 1·1 = 5
```

Output for “book” = 0.75 v(Ali) + 0.04 v(reads) + 0.07 v(a) + 0.13 v(book)

The query of “book” points the same way as the key of “Ali”, so “Ali” gets three quarters of the weight.

WHY DIVIDE BY √D

Without the scale, big vectors make the softmax pick one key

2

spread of q · k at d = 4

8

at d = 64

32

at d = 1,024: the spread grows like √d

Unscaled

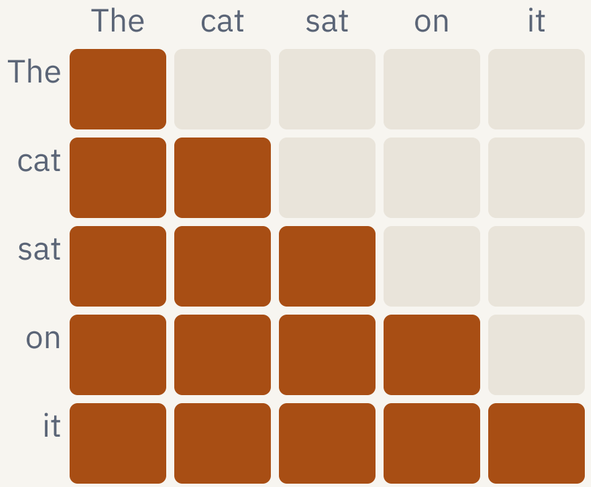
Scores of ±30 make one weight nearly 1 and the rest nearly 0, so the others stop learning

Divided by √d

The spread is back to about 1 at any size. Qwen2.5 uses d = 128 per head

DECODERS

A causal mask stops every token from seeing the future



Dark: may attend. **Light:** blocked.

Row i sees columns 0 … i. Blocked scores are set to −∞ before the softmax, so their weight is exactly 0.

One pass over a sentence of n tokens trains n next-token predictions at once, none of which could copy the answer.

MULTI-HEAD ATTENTION

Many small attentions side by side, each free to track something else



Why split?

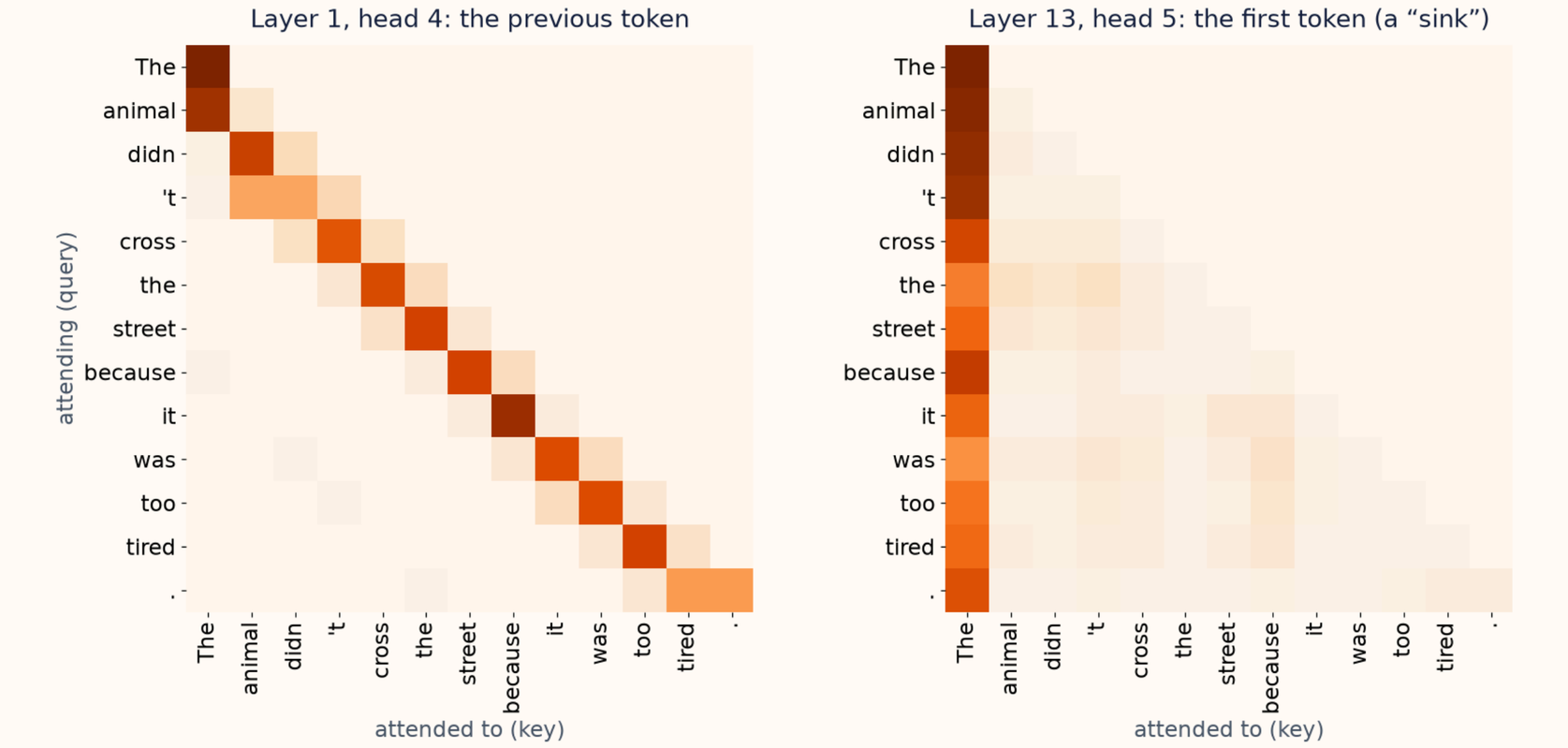
One softmax gives one blend per token. Twelve heads can follow twelve relations, for example the previous word and the subject, for the same cost.

Grouped-query attention

Qwen2.5-1.5B’s 12 query heads share only 2 sets of keys and values: less memory to keep while generating.

INSIDE QWEN2.5-1.5B

Some heads learn one clear job



WORD ORDER

Attention alone cannot tell “dog bites man” from “man bites dog”

2017 · original transformer

Sinusoidal

Add a fixed pattern of sine waves, different for every position, to each embedding

GPT-2, BERT

Learned

Learn one vector per position, but only up to a fixed length, such as 1,024 tokens

Llama, Qwen

Rotary (RoPE)

Rotate each query and key by an angle that grows with position, so q·k depends on the distance between them

Without positions, shuffling the input only shuffles the output. You prove it in Lab 03, Part B.

THE DECODER BLOCK

Attention mixes the tokens; a feed-forward network then works on each one



Attention

The only place where tokens exchange information

Feed-forward

Each token on its own: 1,536 → 8,960 → 1,536 numbers

+ and RMSNorm

Each sub-layer adds to its input; norms keep the numbers stable

Qwen2.5-1.5B stacks **28** of these blocks, then a final norm and the output layer over 151,936 tokens.

WHERE THE PARAMETERS ARE

Attention moves the information, but feed-forward layers hold most of the weights

5.5 M

attention per block (W_Q, W_K, W_V, W_O): 12%

41.3 M

feed-forward per block: 88%

1.54 B

in total: 28 blocks plus the 233 M embedding table

THE COST OF ATTENTION

Every token scores every earlier token, so the work grows with the square of the length

Tokens	Scores per head	All 336 heads
1,000	0.5 million	168 million
8,000	32 million	10.8 billion
32,768	537 million	180 billion

Arabic pays twice

Text that needs 1.5× the tokens needs about 2.25× the attention work. Last week’s tokenizer gap grows here.

TEAM PROJECT · BRIEF RELEASED TODAY

Build and evaluate an LLM application that retrieves and calls a tool

When	Phase	Marks
Week 4	Teams of 3–4 registered	—
Week 7	1 · Problem definition and planning: proposal and data plan	10
Week 10	2 · Design and implementation: dataset, test set and design	10
Weeks 12–15	3 · Testing, evaluation and final submission: prototype, evaluation, final demo	10

Every project

- Answers from documents it retrieves
- Calls at least one tool
- Runs on open-weight models
- Is measured on a test set
- Arabic data preferred

WHAT COMES NEXT

This week: build attention yourself. Next week: training and decoding

Module 2, before the lab

Inside Attention: steer a query, mask the future, reorder a sentence. About 30 minutes; code on Moodle.

Lab 03

Attention from scratch in PyTorch, then the heads inside Qwen2.5, in English and Arabic.

Team and reading

Register your team by the week 4 lab. Read Raschka, chapters 4–5.

Exit question: in one sentence each, what do a token’s query, key and value do?